

File Permission

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drwxrwxr-x 2 ubuntu ubuntu 4096 Dec 3 12:13 demo

- d represent the directory
- rwx ----- user (owner) (jis ne file create ki he)
- rwx ----- group ()
- r-x ----- other user
- 2 ---- represent the link count
- ubuntu ----- represent the owner of the
- ubuntu ----- represent the group owner
- 4096 ----- represent the size
- Dec 3 12:13 ----- date of creation
- Demo ----- name

Directory

- rwx permission (read - ls) (write - mkdir, touch, cp, mv, rm) (excited - cd "without using this command write are not work")

File

- rwx permission (read - cat, tail, head, more) (write - vim, >, >>) (excited - executable file, run)
- u + rwx
- g - wx
- o = r

r w x
4 2 1 = 7

- 0 no permission**
- 1 execute only**
- 2 write only**
- 3 write and execute**
- 4 read only**
- 5 read and execute**
- 6 read and write**
- 7 read write execute**

root user directory 755 file 644
local user directory 775 file 664

Type of file

- User define file
 - Normal file /regular file (-)
 - Directory (d)
 - Link file (l)
- System define file
 - Socket file (s)
 - Pipe file (f)
 - Block device file (b)
 - Character device file (c)

1. chmod
 - Change the permission of file or directory
 -
 -
2. sudo chmod u-w user_name
3. sudo chmod u+w user_name
4. sudo chmod u-rx user_name
5. sudo chmod u-rx user_name
6. cd /home
7. sudo chmod u-rx om
8. ls -l
9. sudo chmod u-w om
10. ls -l
11. sudo chmod r-x om
12. chmod --help
13. chmod -rwx om
14. sudo chmod u=rwx om

15. ls -l
16. sudo chmod o+r om
17. ls -a om

```
ubuntu@ip-172-31-8-114:/home$ ls -a om
ls: cannot access 'om/..': Permission denied
ls: cannot access 'om/.': Permission denied
ls: cannot access 'om/.bash_logout': Permission denied
ls: cannot access 'om/.profile': Permission denied
ls: cannot access 'om/.bashrc': Permission denied
.  .. .bash_logout .bashrc .profile
```

18. ./om/file.txt
19. sudo chmod u=rwx om
20. ls -l
21. sudo chmod u=rwx om
22. sudo chmod om/file.txt

```
ubuntu@ip-172-31-8-114:/home$ sudo chmod om/file.txt
chmod: missing operand after 'om/file.txt'
```

23. sudo chmod o+rw om

```
ubuntu@ip-172-31-8-114:/home$ sudo chmod o+rw om
ubuntu@ip-172-31-8-114:/home$ ls -l
total 12
drwxr-xrwx  2 om          om          4096 Dec  3 12:38 om
drwxr-x---  5 ubuntu     ubuntu     4096 Dec  3 12:13 ubuntu
drwxr-x---  2 vaishali   vaishali   4096 Dec  3 12:03 vaishali
```

24. ./om/file.txt
25. sudo chmod o+x om

```
ubuntu@ip-172-31-8-114:/home$ sudo chmod o+x om
ubuntu@ip-172-31-8-114:/home$ ls -l
total 12
drwxr-xrwx  2 om          om          4096 Dec  3 12:38 om
drwxr-x---  5 ubuntu     ubuntu     4096 Dec  3 12:13 ubuntu
drwxr-x---  2 vaishali   vaishali   4096 Dec  3 12:03 vaishali
```

26. sudo chmod o-w om

```
ubuntu@ip-172-31-8-114:/home$ sudo chmod o-w om
ubuntu@ip-172-31-8-114:/home$ ls -l
total 12
drwxr-xr-x  2 om          om          4096 Dec  3 12:38 om
drwxr-x---  5 ubuntu     ubuntu     4096 Dec  3 12:13 ubuntu
drwxr-x---  2 vaishali   vaishali   4096 Dec  3 12:03 vaishali
```

28. sudo chgrp ubuntu vaishali

```
ubuntu@ip-172-31-8-114:/home$ sudo chgrp ubuntu vaishali
ubuntu@ip-172-31-8-114:/home$ ls -a
.  ..  om  ubuntu  vaishali
ubuntu@ip-172-31-8-114:/home$ ls -l
total 12
drwxr-xr-x  2 vaishali om          4096 Dec  3 12:38 om
drwxr-x---  5 ubuntu  ubuntu     4096 Dec  3 12:13 ubuntu
drwxr-x---  2 vaishali ubuntu     4096 Dec  3 12:03 vaishali
ubuntu@ip-172-31-8-114:/home$ sudo chown amit:tcs dir1
```

29. sudo chown om:tcs dir1

```
ubuntu@ip-172-31-8-114:/home$ sudo chown om:tcs dir1
ubuntu@ip-172-31-8-114:/home$ ls -l
total 12
drwxr-xr-x  2 om          om          4096 Dec  3 12:38 om
drwxr-x---  5 ubuntu     ubuntu     4096 Dec  3 12:13 ubuntu
drwxr-x---  2 vaishali   ubuntu     4096 Dec  3 12:03 vaishali
```

30. mkdir project && touch file.txt
31. ls -l

```

root@ip-172-31-9-173:~# mkdir demo && touch file.txt
root@ip-172-31-9-173:~# ls -l
total 8
drwxr-xr-x 2 root root 4096 Dec  4 12:18 demo
-rw-r--r-- 1 root root    0 Dec  4 12:18 file.txt
drwx----- 3 root root 4096 Dec  4 11:40 snap
root@ip-172-31-9-173:~# su -ubuntu
Try 'su --help' for more information.
root@ip-172-31-9-173:~# su -ubuntu

```

32. umask

```

ubuntu@ip-172-31-9-173:~$ exit
logout
root@ip-172-31-9-173:~# sudo -i
root@ip-172-31-9-173:~# umask
0022
root@ip-172-31-9-173:~# su - ubuntu
ubuntu@ip-172-31-9-173:~$ umask
0002

```

33. umask

- Default 777 - 022 = 755 (directory) 666 - 022 = 644 (file)
- Change 777 - 543 = 234 (directory) 666 - 543 = 123 (file)
- Change the umask means the change the permission of user group and other

```

root@ip-172-31-9-173:~# umask 543
root@ip-172-31-9-173:~# mkdir cc && touch cc.txt
root@ip-172-31-9-173:~# umask
0543
root@ip-172-31-9-173:~# ls -l
ls-l: command not found
root@ip-172-31-9-173:~# ls -l
total 16
drwxr--r-- 2 root root 4096 Dec  4 12:34 aa
-rw-r--r-- 1 root root    0 Dec  4 12:34 bb.txt
d-w--wxr-- 2 root root 4096 Dec  4 12:40 cc
--w--w-r-- 1 root root    0 Dec  4 12:40 cc.txt
drwxr-xr-x 2 root root 4096 Dec  4 12:18 demo
-rw-r--r-- 1 root root    0 Dec  4 12:18 file.txt
drwx----- 3 root root 4096 Dec  4 11:40 snap
root@ip-172-31-9-173:~#

```

34. chmod u+s /sbin/dmidecode

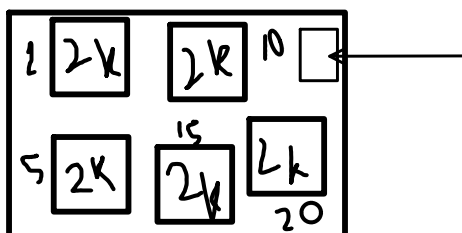
35. su -vaishali

36. chmod u-s /sbin/dmidecode

37.

inode number (series of file data 10k size)

Link file



Difference between Hard Link and Soft Link:

1. Prepose

- Hard link used for backup prepose
- Soft Link are to use creating shortcut

2. Objective

-

Hard Link	Soft Link
Inode number of hard link is same as that of inode number of original file.	Inode number of softlink is different than inode number of original file.
Hardlink contain actual data from the file.	Softlink contain path of original file and not the actual data.
Thus, size of hardlink and original file will be same.	Thus, size of softlink depends on path length.
Creating or removing hardlink will increase or decrease link count by 1.	Creating or removing softlink will not affect link count.
Removing original file will only reduce the link count and it will not affect any of its hardlink.	Removing original file will disable softlink, since softlink points to the non-existing file.
Creating hardlink of directory is not possible.	Creating softlink of directory is possible.
Syntax to create hardlink, # ln <original_file> <hardlink_name>	Syntax to create softlink, # ln -s <original_file> <softlink_name>

3. ln

- Hard link created ln (cc.txt /mnt/hlfile.txt)
- File ke link count se pata kr sakte ho kitni file he

```

root@ip-172-31-9-173:~# cat > cc.txt
hii
hello
wellcome
^Z
[1]+  Stopped                  cat > cc.txt
root@ip-172-31-9-173:~# ln cc.txt /mnt/hlfile.txt
root@ip-172-31-9-173:~# cd /mnt
root@ip-172-31-9-173:/mnt# ls -l
total 4
--w--w-r-- 2 root root 20 Dec  4 13:16 hlfile.txt
root@ip-172-31-9-173:/mnt# cat hlfile.txt
hii
hello
wellcome

```

2. ls -l file.txt

- Show the inode number using this command of the file or directory

1. Viewing File Permissions

- Command:
ls -l
- Example Output:
-rwxr-xr-- 1 user group 1234 Dec 7 10:30 file.txt
 - Breakdown:
 - -rwxr-xr-- → File permissions:
 - - : Regular file (other types: d for directory, l for link).
 - rwx : Permissions for the **owner** (read, write, execute).
 - r-x : Permissions for the **group** (read, execute).
 - r-- : Permissions for **others** (read only).
 - user → File owner.
 - group → Group associated with the file.

2. Understanding Permission Types

- **Read (r)**: View file contents or list directory contents.
- **Write (w)**: Modify file contents or create/delete files in a directory.
- **Execute (x)**: Run a file as a program or access a directory.

3. Changing File Permissions

Using chmod

- Modify permissions for owner (u), group (g), others (o), or all (a).

Symbolic Mode:

- Add permissions:
`chmod u+x file.txt # Add execute for owner`
`chmod g+w file.txt # Add write for group`
`chmod o+r file.txt # Add read for others`
- Remove permissions:
`chmod u-r file.txt # Remove read for owner`
`chmod g-x file.txt # Remove execute for group`
`chmod o-w file.txt # Remove write for others`
- Set exact permissions:
`chmod u=rwx,g=rx,o=r file.txt`

Numeric Mode:

- Use numbers to set permissions:
 - `r = 4, w = 2, x = 1.`
 - Combine values: `rwx = 7, rw- = 6, r-- = 4.`
- Examples:
`chmod 755 file.txt # rwx for owner, rx for group, rx for others`
`chmod 644 file.txt # rw for owner, r for group, r for others`

4. Changing Ownership

Using chown

- Change file owner:
`sudo chown new_owner file.txt`
- Change file group:
`sudo chown :new_group file.txt`
- Change both owner and group:
`sudo chown new_owner:new_group file.txt`

5. Changing Group

Using chgrp

- Command:
`sudo chgrp new_group file.txt`

6. Directory Permissions

- **r**: View directory contents.
- **w**: Create/delete files in the directory.
- **x**: Enter or access the directory.
- Example: Grant full access to a directory and its contents:
`chmod -R 755 /path/to/directory`

7. Special Permissions

Set User ID (SUID):

- Allows a file to run with the owner's privileges.
- Command:
`chmod u+s file`
- Appears as `s` in the owner's execute permission: `-rwsr-xr--`.

Set Group ID (SGID):

- Files created in the directory inherit the group of the directory.
- Command:
`chmod g+s directory`
- Appears as `s` in the group's execute permission: `drwxr-sr-x`.

Sticky Bit:

- Restricts file deletion in a directory to the file's owner.
- Useful for shared directories like `/tmp`.
- Command:
`chmod +t directory`
- Appears as `t` in the others' execute permission: `drwxrwxrwt`.

8. Default Permissions

umask:

- Controls default permissions for new files and directories.
- View current umask:
`umask`
- Example: Set umask to 0022:
`umask 0022`

9. Examples

- Make a script executable for everyone:

`chmod a+x script.sh`

- Restrict access to a file for the owner only:
`chmod 600 secret.txt`
- Allow group members to edit a directory:
`chmod 775 /shared/directory`